# Big Data

BY THE END:

# You can explain what makes data big and know when ordinary tools are no longer enough.

# Overview

A market trader may keep 400 sales in one table, while a digital wallet may record millions of payments every day. Both store transaction data, but the digital wallet needs tools that can handle much more data arriving much faster.

# What makes data big

**Big data** is data that is too large, arrives too quickly, or comes in too many formats for ordinary tools to handle well.

There is no single number that makes data big. What matters is whether the usual tools can still store and process it.

## The three Vs

| PART | SIMPLE QUESTION | MARKET TRADER | DIGITAL WALLET |
|---|---|---|---|
| **Volume** | How much data is there? | 400 sales records | Millions of payment records |
| **Velocity** | How quickly does new data arrive? | A few sales per hour | Thousands of payments per second |
| **Variety** | How many formats does it use? | One sales table | Payment records, app logs, messages, and receipt photos |

A dataset can be difficult to handle because of one V or several at the same time.

# Big depends on the tools

Two organizations can use different tools for the same dataset. A file with 2 million rows may be too large for a spreadsheet but easy for a database.

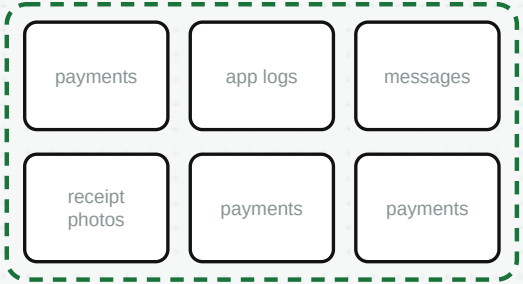
Data becomes big when the tools normally used are no longer enough. Many small shops, schools, and clinics can answer their questions with ordinary tools. They do not need big data systems.

**A market trader**




400 sales records
a few sales per hour

**A digital wallet**




millions of payment records
thousands of payments per second

# What changes when data is big

When one machine is not enough, the work is divided across several machines.

| WHAT CHANGES | WHAT IT MEANS |
| --- | --- |
| **Storage** | Data may be stored across several machines |
| **Processing** | Several machines may work on the data together |
| **Quality** | Missing or incorrect values are harder to find among millions of records |
| **Management** | Copies of data are harder to track, protect, and delete |

# More data is not always better

More data is not always better. A small, clean dataset can be more useful than a large dataset with missing or incorrect values.

The market trader can answer a useful question with 400 accurate sales records. Millions of unreliable records may give the wrong answer.